

Interpretation and Antecedent Biases of Null vs Overt Pronouns in Turkish

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Introduction

Previous studies on anaphora resolution of null and overt subjects suggest that the grammatical function of antecedents plays an important role in resolving anaphora by modulating the **prominence**, **salience**, or **accessibility** of discourse referents.

Position of Antecedent Hypotheses (PAH): Proposed by Carminati (2002), this hypothesis suggests that syntactic function is the primary factor in determining an antecedent in intra-sentential contexts (data from Italian—a *pro*-drop Indo-European language with SVO word order)

- **Null pronouns** → subject antecedents (more prominent)
- **Overt pronouns** → non-subject antecedents (less prominent)

According to PAH, processing difficulty arises when this preference is violated:

(1) *Elif Ayşe'yi azarlayınca, ∅ ağladı*
Elif Ayşe-acc humiliate-when cry-pst.3sg

‘When Elif humiliated Ayşe, she cried.’ (**Biased towards object**)

Research Questions:

1. Do Turkish null and overt subject pronouns exhibit the Position of Antecedent Hypothesis in intra-sentential contexts?
2. Which anaphoric expression (i.e. null pronoun, overt pronoun or R-expression) is preferred for subject and object antecedents in intra-sentential contexts? How does interpretation correlate with production?

Turkish Anaphoric Expressions

Turkish is a *pro*-drop language: subject pronouns can be omitted, as verbs are morphologically marked for person and number. It employs three types of anaphoric expressions:

- a null pronoun (∅)
- an overt pronoun (e.g., *he*, *she*, *it*)
- an R-expression (e.g., *Elif*, *Ayşe*)

Subject Pronouns in Turkish

<i>ben</i>	‘I’	<i>biz</i>	‘we’
<i>sen</i>	‘you(SG)’	<i>siz</i>	‘you(PL)’
<i>o</i>	‘he/she/it’	<i>onlar</i>	‘they’

Null vs Overt Pronoun Subjects

Turkish tends to avoid repeating overt pronouns or R-expressions in subject position (See 2):

(2) *Elif_i yemek ye-di. *Elif_i/*O_i/pro_i çay iç-ti.* → *O_j for topic shift*
Elif/she meal eat-pst tea drink-pst.3sg

‘Elif ate (lit. had a meal). She drank tea.’ Taken from Balkız (2002)

When antecedents are distinguishable via person and number features, null pronouns are preferred (see 3); when not, overt forms are used to clarify antecedent (see 4).

(3) *Ali_i [Filiz’le Ayhan’ı]_j yemeğe çağırdı. pro_j çok meşguldüler.*
Ali Filiz-with Ayhan-acc dinner-dat invite-pst very busy-cop.pst.3pl

‘Ali_k invited Filiz and Ayhan to dinner. They were very busy.’

(4) *∅_i Ali’yi_j ara-dı ama o_j ev-de değil-di.*
Ali-acc call-pst.3sg but he home-loc not-pst

‘(She/He) called Ali, but he was not at home.’ Both taken from Turan (1995)

(4) illustrates a shift-to-object or attention shift, as framed within Centering Theory (Brennan et al., 1987; Turan, 1995). Balkız (2002) similarly classifies such cases as topic shifts and argues that *o_i* is not acceptable in these contexts (see 2).

Turan (1995) reports that in shift-to-object, the **overt pronoun o** occurs in **21%** of instances (27/132), while **R-expressions** (e.g., *Ali* instead of *o*) are more common (**77%** of cases (102/132)).

Experiments

These experiments investigated native Turkish speakers’ (*N* = 23) interpretation of null and overt subject anaphora in intra-sentential contexts. The data were collected via a Google Docs survey

- **Experiment 1** (97 items) tested the predictions of the *Position of Antecedent Hypothesis* (PAH) across three bias conditions with an offline experiment.
- **Experiments 2** (18 items) and **3** (23 items) examined production preferences and naturalness judgments for sentences with null pronouns, overt pronouns, or R-expressions. These explored whether patterns observed in interpretation are reflected in production choices.

Stimuli Examples:

No Bias *Elif Ayşe’yi gördüğünde, (o) yemek yiyordu.*
‘When Elif saw Ayşe, (she) was eating.’

Subject-Biased *Elif Ayşe’yi rezil edince, (o) özür diledi.*
‘When Elif embarrassed Ayşe, (she) apologized.’

Object-Biased *Elif Ayşe’yi azarlayınca, (o) ağladı.*
‘When Elif scolded Ayşe, (she) cried.’

Sample Questions:

- **Exp 1:** Who was eating?
a. Elif b. Ayşe c. Someone else
- **Exp 2:** If Ayşe was eating, which version of the sentence sounds most natural?
a. Elif Ayşe’yi gördüğünde, yemek yiyordu. (null)
b. Elif Ayşe’yi gördüğünde, **o** yemek yiyordu. (overt)
c. Elif Ayşe’yi gördüğünde, **Ayşe** yemek yiyordu. (name)
- **Exp 3:** If Ayşe was eating, how natural does the sentence sound?
“Elif Ayşe’yi gördüğünde, yemek yiyordu.” (null)
1 (Not at all natural) 2 3 4 5 (Very natural)

Methods

Experiment 1:

- In the no-bias condition: Syntactic function significantly affected interpretation of null vs. overt pronouns (see Fig. 1, No bias).
- In the object-biased condition: Null pronouns showed a significant preference for subject antecedents, **supporting PAH** (Fig. 1, Bias towards object).
- In the subject-biased condition: Overt pronouns did not show a strong preference for object antecedents, **not offering strong evidence for PAH** (Fig. 1, Bias towards subject).

Across all bias conditions: The interpretation of overt pronouns as referring to ‘neither’ (i.e., someone else) was significantly higher than that of null pronouns, even in intra-sentential contexts.

Experiment 2: Results were consistent with Experiment 1. In the subject-biased context, overt pronouns were not significantly preferred over null pronouns when referring to object antecedents (Fig. 3, Bias towards subject). The strong preference for R-expressions aligns with Turan (1995).

Experiment 3: Results aligned with Experiment 2. More data is needed for stronger conclusions. **The data was analyzed using logit mixed-effects models and cumulative link mixed models.**

References

Öztürk, B. (2001). Turkish as a Non-Pro-Drop Language. In E. E. and introd. . Taylan (Ed.), *The Verb in Turkish* (pp. 239–259). John Benjamins Publishing Company. Brennan, S. E., Friedman, M. W., & Pollard, C. (1987). A centering approach to pronouns. In *Proceedings of the 25th annual meeting of the Association for Computational Linguistics* (pp. 155–162) Carminati, M. N. (2002). *The processing of Italian subject pronouns*. University of Massachusetts Amherst. Taylan, E. E. (1986). Pronominal versus zero representation of anaphora in Turkish. In *Studies in Turkish linguistics*. John Benjamins. Turan, U. D. (1995). *Null vs. overt subjects in Turkish discourse: A centering analysis*. Pennsylvania: University of Pennsylvania.

Results

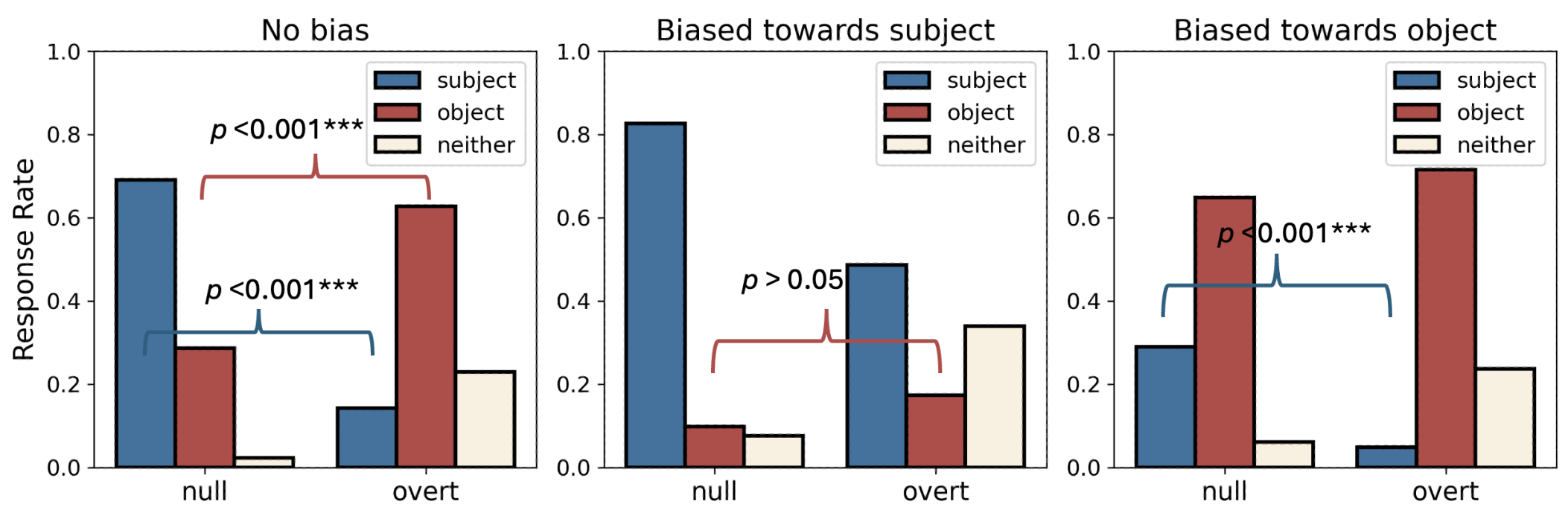


Figure 1. Exp1:Response rates for subject and object antecedents by bias condition and anaphor type.

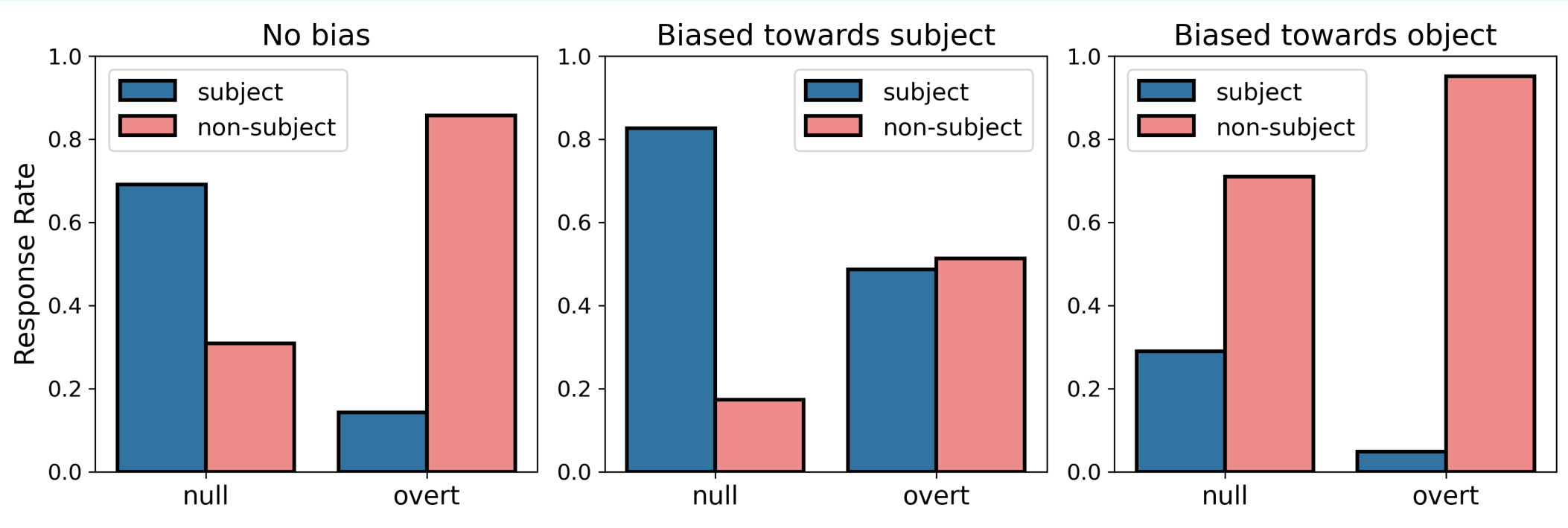


Figure 2. Exp1: Object and Neither responses combined into a single Non-subject category.

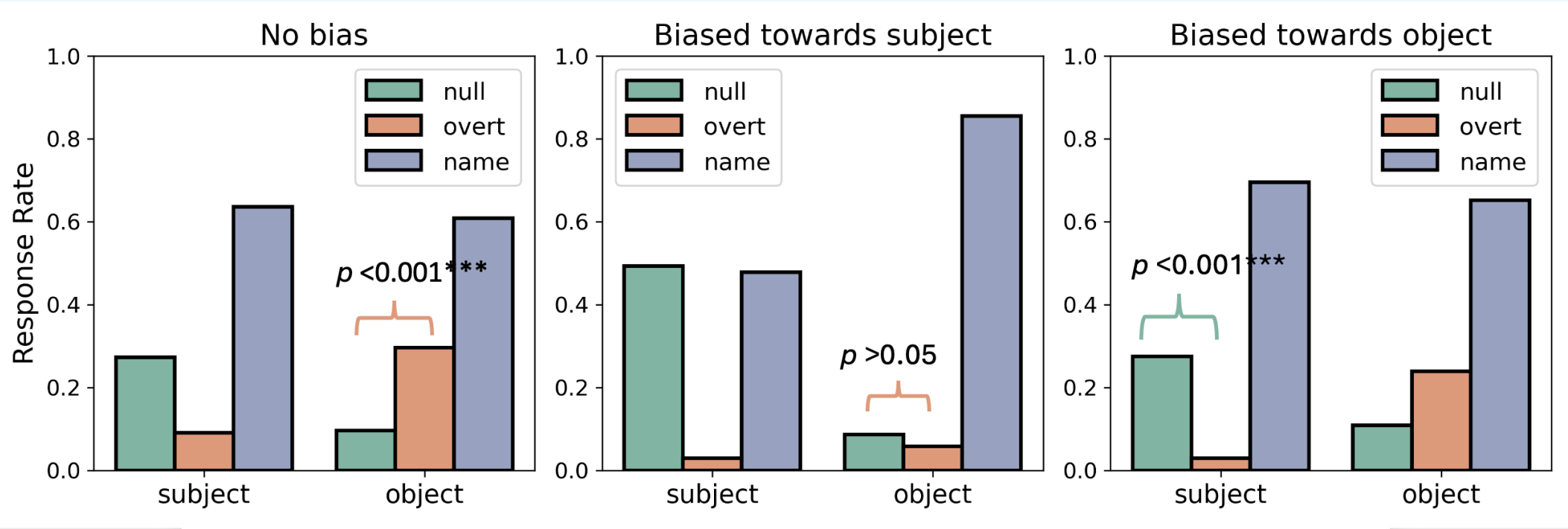


Figure 3. Exp2: Response rates for production preferences by bias condition and antecedent.

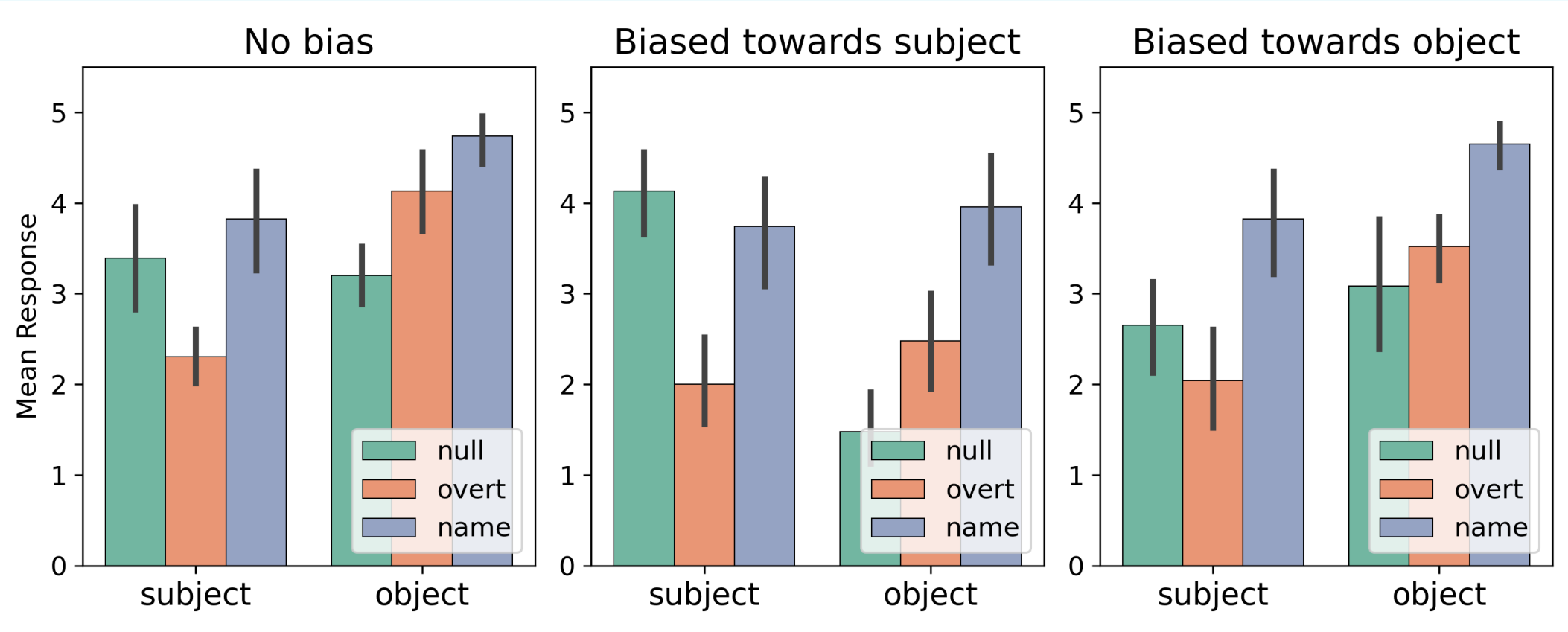


Figure 4. Exp3: Mean responses for naturalness judgments by bias condition, antecedent and anaphor type.

Conclusion and limitations

Note: * $p < .05$; ** $p < .01$; *** $p < .001$

Null pronouns are observed to be more robustly used to refer to subject antecedents across conditions, in line with the predictions of PAH, whereas overt pronouns fail to show a similar preference when the context is biased toward the subject.

This study does not include reading time measures to evaluate processing difficulty. Pragmatically biased items could have stronger cues to better convey the intended bias.